6TH GRADE MATH · UNIT 7: DATA DISPLAYS

# Lunch Menu Survey

*Teacher Guide & Answer Keys*

**Teacher**

2–3 class periods

## Project Overview

Students act as the cafeteria committee. They pose statistical questions, display lunch-line data in dot plots, histograms, and box plots, compute center and spread, and recommend a menu change.

The outlier (12 minutes) gives a natural reason to compare mean and median.

## Standards Addressed

- 6.SP.A.1 Recognize statistical questions
- 6.SP.A.2–3 Describe distributions by center, spread, and shape
- 6.SP.B.4 Display data in dot plots, histograms, and box plots
- 6.SP.B.5 Summarize data in context (mean, median, IQR, MAD)

## Pacing (2–3 class periods)

| Day | Plan |
|---|---|
| Day 1 | Launch: time a real line or use the card. Part 1 displays. |
| Day 2 | Part 2 center and spread; compare two data sets. |
| Day 3 | Part 3 recommendation report. |

## Materials

- Student handout with the Survey Data card
- Grid paper
- Calculators
- Optional: survey your own class

## How the Three Versions Differ

| Level | What changes |
|---|---|
| CP2 | Frequency table, dot plot, mean, median, mode, range, and a bar graph with a simple recommendation. |
| CP1 | Histograms and box plots, IQR outlier rule, MAD, and before/after comparison. |
| Honors | Choosing displays for a purpose, removing outliers, comparing groups in MADs, survey design, and bias. |

## Watch For (Common Misconceptions)

- Finding the median without ordering the data.
- Using unequal intervals in a histogram.
- Treating the bar graph of categories like a histogram.

## Extension Ideas

- Collect real data in your cafeteria.
- Use a spreadsheet to make the displays.
- Survey a second grade level and compare.

# Answer Key – CP2

## Part 1: Ask the Question

| # | Task | Answer |
|---|---|---|
| 1 | Which is a statistical question: "How long did you wait in line today?" or "How long do students wait in the lunch line?" Explain. | The second one; it expects many different answers (variability). |
| 2 | Make a frequency table for the 20 wait times. | 3:1, 4:2, 5:4, 6:6, 7:2, 8:2, 9:1, 10:1, 12:1 |
| 3 | Make a dot plot of the wait times. | Dots above 3–12 matching the frequency table. |
| 4 | What is the mode? What does it tell you? | 6 minutes; it is the most common wait time. |

## Part 2: Center and Spread

| # | Task | Answer |
|---|---|---|
| 5 | Find the median wait time. | Ordered: 10th and 11th values are both 6 → median 6 minutes |
| 6 | Find the mean wait time. | Sum 128 ÷ 20 = 6.4 minutes |
| 7 | Find the range. | 12 − 3 = 9 minutes |
| 8 | Which value is an outlier? How does it change the mean? | 12 is far from the rest; it pulls the mean up a little (without it the mean is 116 ÷ 19 ≈ 6.1). |

## Part 3: Recommend a Menu

| # | Task | Answer |
|---|---|---|
| 9 | Make a bar graph of the favorite foods. | Bars: pizza 7, tacos 5, salad bar 3, burgers 4, other 1. |
| 10 | What fraction of students chose tacos? What percent? | $\frac{5}{20} = \frac{1}{4} = 25\%$ |
| 11 | Which new lunch would you recommend? Use two facts from the data. | Answers vary; e.g., tacos: 25% chose it and pizza is already served; most students wait about 6 minutes, so a quick taco bar fits. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | Why did the outlier change the mean more than the median? | The mean uses every value, so a large value pulls it; the median only depends on the middle. |
| R2 | What makes a question statistical? | It anticipates variability in the answers. |

# Answer Key – CP1

## Part 1: Display the Data

| # | Task | Answer |
|---|---|---|
| 1 | Make a dot plot and describe the shape of the data. | Peak at 6, most values between 4 and 8, a long tail to the right (12) → skewed right. |
| 2 | Make a histogram with intervals 2–3, 4–5, 6–7, 8–9, 10–11, 12–13. | 2–3: 1, 4–5: 6, 6–7: 8, 8–9: 3, 10–11: 1, 12–13: 1 |
| 3 | Find the five-number summary. | Min 3, Q1 5, median 6, Q3 7.5, max 12 |
| 4 | Draw a box plot. Is 12 an outlier by the 1.5 × IQR rule? | IQR = 2.5; upper fence 7.5 + 3.75 = 11.25; 12 > 11.25, so yes, 12 is an outlier. |

## Part 2: Measure It

| # | Task | Answer |
|---|---|---|
| 5 | Find the mean and median. Which better describes a typical wait? Why? | Mean 6.4, median 6; the median, because the data is skewed right by the 12. |
| 6 | Find the mean absolute deviation (MAD). Round to the nearest tenth. | Deviations from 6.4 sum to 32.4 → MAD ≈ 1.6 minutes |
| 7 | After a second register opened, the times were: 3, 4, 4, 5, 5, 5, 4, 6, 3, 4, 5, 5, 4, 6, 5, 4, 3, 5, 4, 6. Find the new median and IQR. | Median 4.5, Q1 4, Q3 5, IQR 1 |
| 8 | Compare the two data sets in two sentences using center and spread. | Typical wait dropped from 6 to 4.5 minutes, and the waits are much more consistent (IQR 2.5 → 1). |

## Part 3: Menu Recommendation

| # | Task | Answer |
|---|---|---|
| 9 | What percent chose each option? Make a display. | Pizza 35%, tacos 25%, salad 15%, burgers 20%, other 5%. |
| 10 | The cafeteria can serve one new lunch only if at least 20% want it and it does not raise the typical wait above 6 minutes. Which options qualify? | Pizza, tacos, and burgers have at least 20%; all qualify on percent. Students argue about wait time. |
| 11 | Write your recommendation with at least one display and two statistics. | Answers vary; look for a display, a measure of center, and a measure of spread. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | When is the median a better summary than the mean? | When the data is skewed or has outliers. |

| # | Prompt | Look for |
|---|---|---|
| R2 | What does a small IQR or MAD tell you about the wait times? | The wait times are close together and predictable. |

# Answer Key – Honors

## Part 1: Choose the Display

| # | Task | Answer |
|---|---|---|
| 1 | Make a box plot and a histogram (intervals of 2 minutes starting at 2). Which display shows the outlier better? Which shows the peak better? | Box plot shows the outlier (12 > 11.25 fence); the histogram shows the peak at 6–7. |
| 2 | Find the mean, median, MAD, and IQR. | Mean 6.4, median 6, MAD ≈ 1.6, IQR 2.5 |
| 3 | If the 12 is removed, which statistics change and by how much? | Mean 116 ÷ 19 ≈ 6.1 (down about 0.3); median 6 (no change); range 9 → 7; IQR: Q1 5, Q3 7 → 2. |
| 4 | A student says, "The mean is 6.4, so most students wait about 6 minutes." Evaluate this statement. | Partly true; 6 is also the median and mode, but the mean alone hides the spread (3 to 12 minutes). |

## Part 2: Compare Lunch Lines

| # | Task | Answer |
|---|---|---|
| 5 | Find the mean, median, and MAD of Line B. | Mean 90 ÷ 20 = 4.5; median 4.5; MAD 0.8 |
| 6 | Draw both box plots on one scale. Describe the overlap. | A: 3, 5, 6, 7.5, 12. B: 3, 4, 4.5, 5, 6. Most of B lies below A's median. |
| 7 | How many MADs apart are the two means (use the larger MAD)? Is the difference meaningful? | (6.4 − 4.5) ÷ 1.6 ≈ 1.2 MADs; a noticeable difference. |
| 8 | Predict the total minutes saved per day for 300 students if the new register is used. Show your reasoning. | (6.4 − 4.5) × 300 = 570 minutes (9.5 hours) per day. |

## Part 3: Survey Design & Recommendation

| # | Task | Answer |
|---|---|---|
| 9 | Write a survey question to choose between tacos and burgers. Describe how to make the sample fair. | Answers vary; ask randomly selected students from every grade and lunch period. |
| 10 | The favorite-food survey (pizza 7, tacos 5, salad 3, burgers 4, other 1) came only from 6th graders. Explain why this might be biased. | It does not represent 7th and 8th graders, whose preferences may differ. |
| 11 | Write a one-page recommendation with at least two displays and three statistics, including a limitation of the data. | Answers vary; look for correct displays, statistics, and a stated limitation. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | How does the purpose of a report affect the display you choose? | Box plots compare spread and outliers; histograms show shape; bar graphs compare categories. |
| R2 | Why should conclusions mention the sample and its limits? | A biased or small sample may not represent the whole school. |